

Project Report On

Seizure Localization Using Quantum Algorithms

Submitted in partial fulfillment of the requirements for the award of the degree of

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in

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By

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CERTIFICATE

This is to certify that the project report entitled ***"Seizure Localization Using Quantum Algorithms"*** *is a bonafide record of the work done by* ***Joyal George Joseph (U2203123)****, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

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Abstract

Epilepsy is a neurological disorder characterized by recurrent seizures caused by abnormal electrical activity in the brain. The unpredictable nature of seizures presents significant challenges for patient safety and quality of life, making early detection of seizure onset an important objective in neurological monitoring systems. This project presents a computational framework for analysing electroencephalography (EEG) signals to identify seizure-related brain states and provide early warning of seizure onset. EEG recordings from the CHB-MIT dataset are first preprocessed and segmented into analysis windows, from which descriptive features capturing temporal and spectral characteristics of neural activity are extracted. These features are used to train machine learning models for distinguishing seizure states. Classical Support Vector Machine (SVM) models are evaluated using different feature combinations, and a hybrid quantum–classical approach using a Quantum Support Vector Machine (QSVM) is also explored to investigate the potential of quantum feature embeddings for EEG analysis. To capture the temporal evolution of seizure activity, a Temporal Convolutional Network (TCN) is implemented to analyse sequences of EEG feature vectors and detect patterns preceding seizure onset. Experimental results demonstrate that the proposed system is capable of identifying seizure states and generating early warnings prior to seizure events. The study highlights the effectiveness of combining signal processing, machine learning, and temporal modelling techniques for EEG-based seizure analysis while also exploring the potential of emerging quantum computing approaches for future biomedical signal processing applications.

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List of Abbreviations

EEG	Electroencephalogram
SVM	Support Vector Machine
QSVM	Quantum Support Vector Machine
TCN	Temporal Convolutional Network
CHB-MIT	Children's Hospital Boston – Massachusetts Institute of Technology EEG Dataset
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
ICA	Independent Component Analysis
QI	Quantum-Inspired
ANOVA	Analysis of Variance
QML	Quantum Machine Learning
EDF	European Data Format
PSD	Power Spectral Density

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Chapter 1

Introduction

This chapter introduces the problem of epileptic seizure analysis and the motivation behind developing computational approaches for seizure localisation and early detection. The study explores how electroencephalography (EEG) signals can be analysed using classical and quantum-inspired machine learning methods to better understand seizure dynamics and support early warning systems.

1.1 Background

A seizure is a sudden burst of abnormal electrical activity in the brain that temporarily disrupts normal neuronal communication and can affect movement, awareness, or behaviour. Epilepsy is a chronic neurological disorder in which a person experiences recurrent, unprovoked seizures due to abnormal brain activity. Epilepsy affects approximately 50 million people worldwide, making it one of the most common neurological disorders [1].

Electroencephalography (EEG) is a primary clinical tool used to record the electrical activity of the brain. EEG signals reflect the collective activity of neurons and allow clinicians to observe abnormal patterns associated with seizures [2]. Seizure-related brain activity progresses through several stages, including interictal, preictal, ictal, and postictal phases. However, the transition between these stages is often gradual and difficult to detect due to the presence of noise and variability in EEG recordings [3].

1.2 Problem Definition

Seizure-related brain activity progresses through multiple stages with complex and inconsistent transitions, making accurate time-based localisation of seizures difficult [3]. This project aims to explore computational methods, including classical machine learning and hybrid quantum approaches, to analyse EEG signals and improve seizure localisation and

early detection.

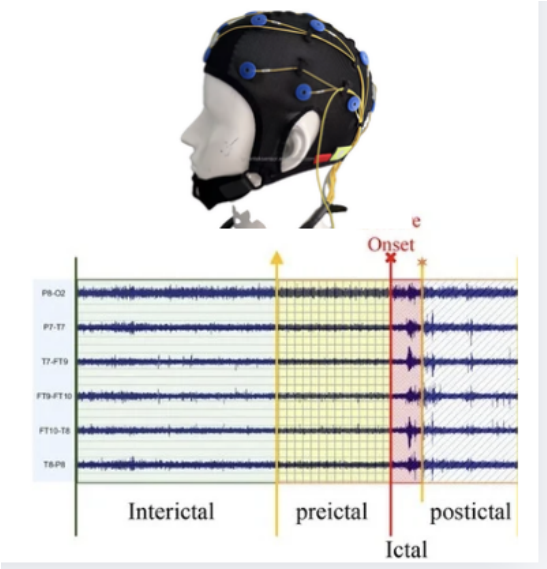


Figure 1.1: Illustrative representation of EEG-based seizure analysis.

1.3 Scope and Motivation

The scope of this project is to study EEG signals and analyse seizure-related brain activity using advanced computational techniques. The system processes EEG recordings by performing preprocessing, feature extraction, and machine learning-based modelling. The study investigates time-domain, frequency-domain, and quantum-inspired features extracted from EEG data. Both classical models such as Support Vector Machines and hybrid quantum–classical approaches are explored. Additionally, a Temporal Convolutional Network (TCN) framework is developed to capture temporal patterns in EEG signals and analyse the transition between seizure stages.

The motivation for this work arises from the unpredictable nature of epileptic seizures and the risks they pose to patient safety. Seizures often occur suddenly, leading to injuries and serious health complications. Although EEG signals contain subtle patterns that appear before seizures, these patterns are often hidden within noisy recordings and are difficult to detect [4, 5]. Developing computational methods capable of identifying early changes in brain activity could enable early warning systems and allow timely medical intervention. Improving seizure detection and localisation can significantly enhance

patient monitoring and safety.

Table 1.1: Major Seizure Stages Observed in EEG Signals

Stage	Description	Significance
Interictal	Normal brain activity between seizures	Baseline EEG activity
Preictal	Transition period before seizure onset	Important for early prediction
Ictal	Period during the seizure	Abnormal brain activity
Postictal	Recovery phase after seizure	Brain returns to baseline

1.4 Objectives

- To study EEG activity across different stages of epileptic seizures.
- To identify and analyse important EEG features related to seizure stages.
- To develop computational models for seizure localisation using extracted features.
- To explore hybrid quantum–classical approaches for modelling complex EEG signals.
- To evaluate the effectiveness of the proposed models in detecting seizure dynamics.

1.5 Challenges

EEG recordings often contain noise, artifacts, and variability that can affect the reliability of feature extraction and model predictions [2, 3]. Additionally, the transition between seizure stages is gradual and complex, making accurate detection and localisation difficult.

1.6 Assumptions

- EEG recordings accurately represent the electrical activity of the brain.
- Preprocessing techniques effectively remove noise and artifacts from EEG signals.
- Extracted features are sufficient to represent seizure-related brain dynamics.

1.7 Societal / Industrial Relevance

Early detection and localisation of epileptic seizures can significantly improve patient safety and healthcare monitoring. Systems capable of analysing EEG signals in real time could assist clinicians in diagnosis and enable automated patient monitoring systems. In healthcare industries, such systems may support the development of intelligent medical devices and wearable technologies for seizure monitoring and early warning applications [4, 5].

1.8 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 presents the literature review and discusses existing research related to EEG-based seizure detection and analysis. Chapter 3 describes the feature engineering framework, including dataset description, EEG signal preprocessing, and extraction of time-domain, frequency-domain, and quantum-inspired features. Chapter 4 presents the modelling methodology, including classical machine learning models, hybrid quantum–classical approaches, and temporal modelling using a Temporal Convolutional Network. Chapter 5 presents the experimental results and discussion of the proposed methods. Finally, Chapter 6 summarizes the conclusions of the study and outlines directions for future work.

1.9 Summary of Chapter

This chapter introduced the background and significance of epileptic seizure analysis using EEG signals. The problem definition, scope, motivation, and objectives of the project were discussed. Challenges, assumptions, and the societal relevance of the proposed system were also highlighted. The next chapter reviews existing research related to seizure detection and EEG analysis techniques.

Chapter 2

Literature Survey

This chapter reviews existing research related to epileptic seizure detection, EEG signal analysis, and machine learning approaches used for identifying seizure patterns. Several studies have explored different computational techniques including signal processing, classical machine learning, and recent deep learning approaches for analysing EEG data. This chapter summarizes important contributions in this area and identifies the limitations of existing approaches that motivate the proposed work.

2.1 Machine Learning for Seizure Detection

Shoeb and Guttag proposed one of the early machine learning based approaches for epileptic seizure detection using EEG signals. Their work focused on extracting discriminative features from EEG recordings and training machine learning classifiers to detect seizure onset. The study demonstrated that automated machine learning systems can assist clinicians in identifying seizure patterns more efficiently. However, the system required patient-specific training and relied heavily on handcrafted features extracted from EEG signals [6].

Shoeb further extended this work in his doctoral research by developing advanced machine learning models for seizure onset detection. The study utilized EEG datasets containing seizure recordings and explored different feature extraction methods and classification techniques. The results showed promising performance for automated seizure detection systems. Nevertheless, the approach still depended on carefully engineered features and did not fully address variability across patients [7].

2.2 EEG Signal Analysis Techniques

Acharya et al. conducted a comprehensive review of automated EEG analysis methods used for epilepsy diagnosis. The authors discussed several signal processing and machine learning techniques for detecting epileptic seizures. The study highlighted the importance of time-domain and frequency-domain features in representing EEG signals. It also emphasized the role of computational intelligence in assisting clinical diagnosis. Despite the progress made, the review pointed out that EEG data are highly complex, noisy, and difficult to interpret, which poses challenges for automated systems [8].

Hjorth introduced time-domain parameters for EEG analysis, which are widely used for characterizing brain signal dynamics. The Hjorth parameters, namely activity, mobility, and complexity, provide quantitative measures of signal characteristics and have been widely applied in biomedical signal processing. Although these features are computationally efficient, they may not fully capture complex spectral characteristics present in EEG signals [9].

2.3 Time-Frequency and Advanced Computational Methods

Cohen introduced time–frequency analysis techniques that enable simultaneous representation of signals in both time and frequency domains. These methods are particularly useful for analysing non-stationary signals such as EEG recordings. Time–frequency approaches provide richer information about signal behaviour and are widely used for seizure analysis. However, they may introduce higher computational complexity and require careful parameter selection [10].

Welch proposed the power spectral density estimation method using the Fast Fourier Transform (FFT). The Welch method is widely used in EEG analysis for estimating signal power across different frequency bands. This approach improves spectral estimation accuracy compared to simple FFT-based methods. However, it still assumes stationary signal behaviour within each analysis window [11].

Recent research has also explored quantum machine learning approaches for analysing complex datasets. Schuld et al. introduced the concept of quantum machine learning and discussed how quantum computing techniques could enhance pattern recognition and optimisation tasks. Although these methods are still emerging, they show potential for

modelling complex biomedical data such as EEG signals [12].

2.4 Summary and Gaps Identified

Summary of Literature

Table 2.1: Comparison of Existing Literature

Author	**Method Used**	**Advantages**	**Limitations**
Shoeb & Guttag (2010)	Machine learning based seizure detection	Automated detection using EEG features	Requires patient specific training
Shoeb (2009)	Feature based seizure onset detection	High detection accuracy using EEG features	Relies on handcrafted features
Acharya et al. (2013)	Review of EEG analysis techniques	Comprehensive overview of automated EEG analysis	EEG complexity and noise remain major challenges
Hjorth (1970)	Time domain EEG parameters	Simple and computationally efficient features	Limited representation of complex EEG patterns
Cohen (1995)	Time–frequency signal analysis	Captures both temporal and spectral signal characteristics	Computationally expensive
Welch (1967)	Power spectral density estimation	Accurate frequency domain representation	Assumes signal stationarity
Schuld et al. (2015)	Quantum machine learning	Potential for modelling complex data	Still in early research stage

Gaps Identified

1. Many existing seizure detection systems rely heavily on handcrafted features, which may not fully capture the complex patterns present in EEG signals.

2. Traditional machine learning models often require patient-specific training, limiting their ability to generalize across different patients.

3. EEG signals are highly noisy and non-stationary, making accurate feature extraction and classification difficult.

4. Most existing approaches focus on seizure detection rather than early seizure prediction or localisation across temporal stages.

5. Emerging techniques such as quantum machine learning have not been extensively explored for EEG-based seizure analysis.

2.4.1 Research Gaps Addressed in This Work

This work addresses a subset of the research gaps identified in the literature through the development of an EEG-based seizure analysis framework.

To broaden the range of feature representations used for EEG analysis, multiple feature domains were investigated, including time-domain, frequency-domain, and quantum-inspired features. The inclusion of quantum-inspired descriptors enabled exploration of alternative structural representations of EEG signals beyond conventional statistical features.

To reduce dependence on patient-specific models, a patient-independent classification framework was implemented to distinguish between preictal and ictal brain states. In addition, the potential of quantum-based machine learning techniques was examined through a hybrid quantum–classical model using a Quantum Support Vector Machine.

Finally, to move beyond simple seizure detection and incorporate temporal dynamics, sequential EEG feature vectors were analysed using a Temporal Convolutional Network, enabling modelling of seizure progression and generation of early warnings prior to seizure onset.

2.5 Summary of Chapter

This chapter reviewed important literature related to EEG signal analysis and epileptic seizure detection using computational techniques. Various methods including machine learning, signal processing, and time-frequency analysis were discussed. A comparison of existing approaches highlighted their strengths and limitations. The identified research gaps motivate the development of improved models that combine advanced feature extraction, temporal modelling, and hybrid quantum-classical learning approaches for better seizure localisation and early warning systems.

Chapter 3

Feature Engineering

3.1 Dataset Description

This work utilizes the CHB-MIT Scalp EEG Database [6], a widely used dataset for epileptic seizure analysis. The dataset contains scalp electroencephalography (EEG) recordings collected from 23 pediatric patients undergoing clinical monitoring for epilepsy. Recordings were acquired using electrodes placed according to the international 10–20 electrode placement system, allowing neural activity to be measured from multiple spatial locations across the scalp. The spatial arrangement of electrodes in the 10–20 system is illustrated in Figure 3.1.

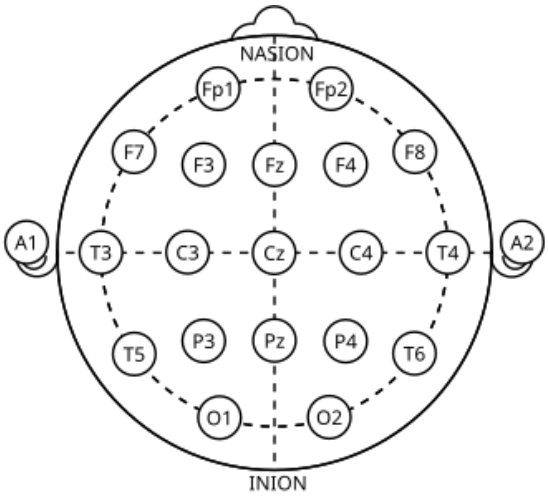


Figure 3.1: Spatial arrangement of electrodes in the international 10–20 EEG electrode placement system.

Each recording consists of multichannel EEG signals stored in European Data Format (EDF), typically containing around 23 channels sampled at 256 Hz. The dataset contains long-duration monitoring sessions spanning several hours and includes 198 annotated

seizure events. These annotations enable the identification of EEG segments corresponding to seizure and non-seizure brain activity used for subsequent signal processing and feature extraction.

3.2 EEG Signal Preprocessing

Raw EEG recordings frequently contain artifacts and noise originating from physiological activity, measurement hardware, and environmental interference. Common artifacts include muscle activity, eye movements, electrode drift, and electrical power-line noise. In order to ensure reliable feature extraction, preprocessing techniques are applied to enhance signal quality and isolate physiologically meaningful components of neural activity.

Signal conditioning procedures are used to remove unwanted frequency components from the recordings. Band-pass filtering is applied to suppress slow baseline drift and high-frequency noise outside the typical EEG frequency range [13]. In addition, notch filtering is used to eliminate narrow-band interference caused by electrical power lines. Artifact removal methods such as Independent Component Analysis (ICA) may also be employed to separate neural signals from artifacts caused by eye blinks and muscle activity.

3.3 Feature Engineering Overview

Electroencephalography signals are high-dimensional time-series recordings that contain large volumes of temporal and spatial information. Directly applying machine learning models to raw EEG signals is computationally inefficient and often fails to capture meaningful patterns due to noise and redundancy in the data. Feature engineering is therefore employed to transform the raw signals into numerical descriptors that summarize relevant characteristics of neural activity.

In this work, features are extracted from three complementary domains:

- **Time-Domain Features:** Statistical descriptors that characterize the amplitude and distribution of the EEG waveform.
- **Frequency-Domain Features:** Spectral descriptors that represent how signal energy is distributed across EEG frequency bands.

- **Quantum-Inspired Features:** Structural descriptors derived from quantum information theory that capture correlations and structural complexity within the EEG signal.

The base descriptors described in the following sections are computed independently for each EEG channel. Statistical aggregation operations such as the mean and standard deviation are then applied across channels to produce channel-independent representations.

3.3.1 Time-Domain Features

Time-domain features describe statistical characteristics of EEG signals directly from the temporal waveform. These features capture amplitude variations, signal variability, and waveform irregularities that may arise during epileptic activity.

Mean Amplitude : The mean amplitude represents the average value of the EEG signal within a time window. It provides an estimate of the baseline electrical activity of the brain. The mean amplitude is defined by Equation 3.1 [14].

$$\bar{x} = \frac{1}{N} \sum_{i=1}^{N} x_i \tag{3.1}$$

In Equation 3.1, $\bar{x}$ denotes the mean amplitude, x_i represents the signal amplitude at sample i, and N represents the total number of samples in the time window. The mean amplitude provides a measure of the central tendency of the signal. Changes in baseline amplitude may reflect shifts in neural activity that occur during abnormal brain states preceding seizure onset.

Standard Deviation : Standard deviation measures the variability of the EEG signal around its mean value. It quantifies how widely the signal amplitudes deviate from the average value. The standard deviation is defined by Equation 3.2 [14].

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (x_i - \mu)^2} \tag{3.2}$$

In Equation 3.2, σ represents the standard deviation of the signal within the time window, x_i denotes the signal amplitude at sample i, μ represents the mean amplitude

defined in Equation 3.1, and N represents the total number of samples in the time window. Larger values of σ indicate greater variability in the EEG waveform. Increased variability is often observed during epileptic seizures due to irregular neural firing.

Skewness : Skewness measures the asymmetry of the signal amplitude distribution around its mean. It is defined by Equation 3.3 [15].

$$\gamma_1 = \frac{E[(x-\mu)^3]}{\sigma^3} \tag{3.3}$$

In Equation 3.3, γ_1 represents the skewness coefficient, x denotes the signal amplitude within the time window, μ represents the mean amplitude of the signal, σ represents the standard deviation, and $E[\cdot]$ denotes the expected value operator. Large skewness values indicate asymmetric waveform distributions, which may arise from abnormal electrical discharges in the brain.

Kurtosis : Kurtosis measures the peakedness of the signal distribution relative to a normal distribution. It is defined using Equation 3.4 [15].

$$\gamma_2 = \frac{E[(x-\mu)^4]}{\sigma^4} - 3 \tag{3.4}$$

In Equation 3.4, γ_2 represents the kurtosis coefficient, x denotes the signal amplitude, μ represents the mean amplitude of the signal, σ represents the standard deviation, and $E[\cdot]$ denotes the expected value operator. High kurtosis values indicate the presence of sharp peaks or spikes in the signal. Such spikes are characteristic of epileptic discharges observed during seizures.

Zero Crossing Rate : The zero crossing rate measures how frequently the signal changes sign within a time window. It is defined by Equation 3.5 [16].

$$Z = \frac{1}{N-1} \sum_{i=1}^{N-1} \mathbf{1}_{\{x_i x_{i+1} < 0\}} \tag{3.5}$$

In Equation 3.5, Z denotes the zero crossing rate, x_i and x_{i+1} represent the signal amplitudes at consecutive samples i and $i+1$, N denotes the total number of samples in the time window, and $\mathbf{1}_{\{\cdot\}}$ represents an indicator function that equals 1 when the

condition is satisfied and 0 otherwise. Higher zero crossing rates indicate rapid oscillatory activity in the signal.

Hjorth Parameters : Hjorth parameters describe the statistical properties of EEG signals in the time domain [9].

Hjorth Activity : measures the signal variance and represents the overall signal power.

$$A = \sigma^2(x) \tag{3.6}$$

In Equation 3.6, A represents the Hjorth activity parameter, $\sigma^2(x)$ denotes the variance of the EEG signal x within the analysis window, and x represents the EEG signal amplitude sequence. It represents the total energy contained within the signal segment.

Hjorth Mobility : measures the mean frequency of the signal.

$$M = \sqrt{\frac{\sigma^2(\dot{x})}{\sigma^2(x)}} \tag{3.7}$$

In Equation 3.7, M represents Hjorth mobility, $\sigma^2(\dot{x})$ denotes the variance of the first derivative of the signal, $\sigma^2(x)$ denotes the variance of the signal, and $\dot{x}$ represents the first temporal derivative of the EEG signal x. It represents the first derivative of the signal.

Hjorth Complexity : measures the variation in frequency components of the EEG signal and reflects the rate of change of the signal frequency.

$$C = \frac{M(\dot{x})}{M(x)} \tag{3.8}$$

In Equation 3.8, C represents the Hjorth complexity parameter, $M(x)$ denotes the Hjorth mobility of the EEG signal x, $M(\dot{x})$ represents the Hjorth mobility of the first derivative of the signal, x denotes the EEG signal amplitude sequence within the analysis window, and $\dot{x}$ represents the first temporal derivative of the signal.Changes in Hjorth parameters may indicate evolving neural dynamics associated with seizure activity.

3.3.2 Frequency-Domain Features

Frequency-domain features describe EEG signals based on their spectral energy distribution.

Power Spectral Density (Welch Method) : Power Spectral Density (PSD) describes how the power of a signal is distributed across different frequencies. In this work, the PSD is estimated using Welch's method, which improves spectral estimation stability by averaging periodograms computed from overlapping signal segments [11]. The Welch PSD estimator is defined in Equation 3.9.

$$S_{xx}(f) = \frac{1}{K} \sum_{k=1}^{K} P_k(f) \tag{3.9}$$

In Equation 3.9, $S_{xx}(f)$ represents the estimated power spectral density at frequency f, K denotes the number of segmented windows used in the estimation, and $P_k(f)$ represents the periodogram of the kth windowed segment of the signal. Each periodogram $P_k(f)$ is computed from the Fourier transform of the windowed signal segment.

Welch's method reduces variance in spectral estimation by averaging the power spectra of multiple overlapping segments. This approach produces a smoother and more reliable estimate of the frequency distribution of EEG signals, which is important for analyzing changes in spectral energy associated with epileptic seizure activity.

Band Power : Band power measures the total spectral energy contained within a specific EEG frequency band. It is defined by Equation 3.10.

$$P_b = \int_{f_{low}}^{f_{high}} S(f) df \tag{3.10}$$

In Equation 3.10, P_b represents the total spectral power within a specific frequency band, $S(f)$ denotes the power spectral density at frequency f, and f_{low} and f_{high} represent the lower and upper frequency limits of the selected EEG band. Changes in band power may indicate abnormal neural oscillations associated with seizure activity.

3.3.3 Quantum-Inspired Features

Quantum-inspired (QI) features are derived from concepts in quantum information theory and are used to capture structural correlations within EEG signals. In this approach, the feature vector extracted from the EEG signal is interpreted as a quantum state representation. This allows structural properties of the signal to be analysed using measures originally developed for quantum systems.

Table 3.1: EEG Frequency Bands and Their Significance

Band	Frequency Range	Significance
Delta	0.5–4 Hz	Associated with deep sleep and pathological states. Increased delta power often appears during seizures.
Theta	4–8 Hz	Associated with drowsiness and memory processes.
Alpha	8–13 Hz	Associated with relaxed wakefulness.
Beta	13–30 Hz	Associated with active cognitive processing.
Gamma	30–40 Hz	Associated with high-frequency neural synchronization.

Density Matrix Representation : In the proposed methodology, the EEG feature vector obtained from the common channels is transformed into a density matrix representation, as defined in Equation 3.11 [17, 18].

$$\rho = \frac{xx^T}{Tr(xx^T)} \tag{3.11}$$

In Equation 3.11, x represents the feature vector extracted from the EEG segment, x^T denotes its transpose, and $Tr(\cdot)$ represents the trace operator that computes the sum of diagonal elements of a matrix. The matrix ρ is normalized such that its trace equals one. This density matrix representation captures correlation relationships between signal components and forms the basis for computing quantum-inspired descriptors.

Von Neumann Entropy : Von Neumann entropy measures the uncertainty associated with a quantum state and is defined by Equation 3.12 [17].

$$S(\rho) = -Tr(\rho \log \rho) \tag{3.12}$$

In Equation 3.12, ρ represents the density matrix defined in Equation 3.11, log denotes the matrix logarithm, and $Tr(\cdot)$ represents the trace operator. Larger values of $S(\rho)$ indicate greater disorder or complexity in the signal structure, which may correspond to

irregular neural activity during seizure states.

Purity : Purity measures how close a quantum state is to a pure state and is defined using Equation 3.13 [17].

$$Purity = Tr(\rho^2) \tag{3.13}$$

In Equation 3.13, ρ denotes the density matrix and ρ^2 represents the matrix product of ρ with itself. The trace operator computes the sum of diagonal elements of the matrix. Higher purity values indicate stronger structural organization within the signal representation.

Linear Entropy : Linear entropy provides a simplified measure of the mixedness of a quantum state and is defined by Equation 3.14 [17].

$$S_L = 1 - Tr(\rho^2) \tag{3.14}$$

In Equation 3.14, ρ represents the density matrix and $Tr(\rho^2)$ corresponds to the purity defined in Equation 3.13. Linear entropy therefore measures the level of disorder within the signal representation.

Quantum Coherence : Quantum coherence quantifies correlations between components of a quantum state and is defined using the l_1 norm of coherence in Equation 3.15 [19].

$$C_{l1}(\rho) = \sum_{i \neq j} |\rho_{ij}| \tag{3.15}$$

In Equation 3.15, ρ_{ij} represents the elements of the density matrix ρ, and the summation is taken over all off-diagonal elements where $i \neq j$. Larger coherence values indicate stronger correlations between signal components.

Quantum Fisher Information : Quantum Fisher Information measures the sensitivity of a quantum state to parameter variations and is defined in Equation 3.16 [20].

$$F_Q = Tr(\rho L^2) \tag{3.16}$$

In Equation 3.16, F_Q denotes the quantum Fisher information, ρ represents the density matrix defined in Equation 3.11, L denotes the symmetric logarithmic derivative operator associated with the parameter being estimated, and $Tr(\cdot)$ represents the trace operator. Higher Fisher information indicates greater sensitivity to changes in signal structure.

Quantum Mutual Information : Quantum mutual information measures the total correlation between subsystems of a quantum system and is defined by Equation 3.17 [17].

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \tag{3.17}$$

In Equation 3.17, ρ_{AB} represents the joint density matrix of the system, while ρ_A and ρ_B represent reduced density matrices of subsystems obtained through partial trace operations.

Entanglement Entropy : Entanglement entropy quantifies the degree of correlation between subsystems of a quantum state and is defined in Equation 3.18 [17].

$$S_E = -Tr(\rho_A \log \rho_A) \tag{3.18}$$

In Equation 3.18, ρ_A denotes the reduced density matrix obtained from the global density matrix ρ. This measure captures dependencies between partitions of the signal representation.

Eigenvalue Mean : The mean of the eigenvalues of the density matrix provides a summary of the spectral distribution and is defined in Equation 3.19 [18].

$$\bar{\lambda} = \frac{1}{n} \sum_{i=1}^{n} \lambda_i \tag{3.19}$$

In Equation 3.19, $\bar{\lambda}$ represents the mean of the eigenvalues of the density matrix ρ, λ_i denotes the ith eigenvalue of ρ, n represents the total number of eigenvalues, and i denotes the index used in the summation.

Eigenvalue Standard Deviation : The spread of the eigenvalue distribution is measured using the standard deviation defined in Equation 3.20 [18].

$$\sigma_\lambda = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(\lambda_i - \bar{\lambda})^2} \tag{3.20}$$

In Equation 3.20, σ_λ represents the standard deviation of the eigenvalue distribution, λ_i denotes the ith eigenvalue of the density matrix ρ, $\bar{\lambda}$ represents the mean eigenvalue defined in Equation 3.19, n denotes the total number of eigenvalues, and i represents the index of the eigenvalue in the summation.

Eigenvalue Skewness : Eigenvalue skewness measures asymmetry of the eigenvalue distribution and is defined in Equation 3.21 [15].

$$\gamma_1 = \frac{E[(\lambda - \bar{\lambda})^3]}{\sigma_\lambda^3} \tag{3.21}$$

In Equation 3.21, λ represents eigenvalues of the density matrix ρ, $\bar{\lambda}$ represents the mean eigenvalue defined in Equation 3.19, σ_λ represents the standard deviation defined in Equation 3.20, and $E[\cdot]$ denotes the expected value operator.

Eigenvalue Kurtosis : Eigenvalue kurtosis measures the peakedness of the eigenvalue distribution and is defined by Equation 3.22 [15].

$$\gamma_2 = \frac{E[(\lambda - \bar{\lambda})^4]}{\sigma_\lambda^4} - 3 \tag{3.22}$$

In Equation 3.22, λ represents eigenvalues of the density matrix ρ, $\bar{\lambda}$ represents the mean eigenvalue, σ_λ represents the eigenvalue standard deviation, and $E[\cdot]$ denotes the expected value operator. Higher kurtosis values indicate concentration of energy in a small number of eigencomponents, which may correspond to dominant structural patterns in the EEG signal.

3.4 Summary

This chapter described the signal processing and feature engineering framework used to represent EEG recordings. Time-domain and frequency-domain descriptors capture the statistical and spectral characteristics of EEG signals, while quantum-inspired features provide additional structural information derived from density-matrix representations.

These descriptors are extracted from individual EEG channels and subsequently aggregated to form the final feature representation used in this work.

Chapter 4

System Design

This chapter describes the design of the proposed EEG-based seizure state classification and prediction system. The system integrates signal processing, feature engineering, classical machine learning, and quantum-inspired modelling techniques to analyse EEG recordings and identify seizure-related patterns [3, 2].

The design evolved progressively through two modelling stages. The first stage focuses on a controlled classification problem that distinguishes between seizure-related EEG states using engineered features. This stage enables the effectiveness of different feature representations and modelling strategies to be evaluated. The second stage extends the system toward temporal EEG modelling, where sequences of EEG features are analysed to detect seizure progression directly from continuous recordings.

4.1 Overall System Architecture

The proposed system follows a structured processing pipeline for EEG analysis. At a high level, the system processes EEG recordings through a sequence of stages including signal preprocessing, feature extraction, model development, and performance evaluation.

Raw EEG recordings are first subjected to signal conditioning in order to remove noise and artifacts. The preprocessed signals are then segmented into analysis windows from which descriptive features are computed. These features capture statistical characteristics, spectral properties, and structural relationships present in the EEG signals [3].

The resulting feature representations are used as input to machine learning models that perform seizure analysis. Two complementary modelling approaches are implemented within the system. The first approach focuses on feature-based seizure state classification using classical and quantum-inspired classifiers. The second approach extends the framework toward temporal EEG modelling, allowing seizure progression and early seizure

warning to be identified directly from continuous EEG recordings.

4.2 System Design Diagrams

The overall processing structure of the EEG analysis system is illustrated using the architecture diagrams presented in Figures 4.1 and 4.2. These diagrams summarize the interaction between the major components of the system.

4.2.1 Feature-Based Classification Pipeline

Figure 4.1 illustrates the processing pipeline used for feature-based seizure state classification.

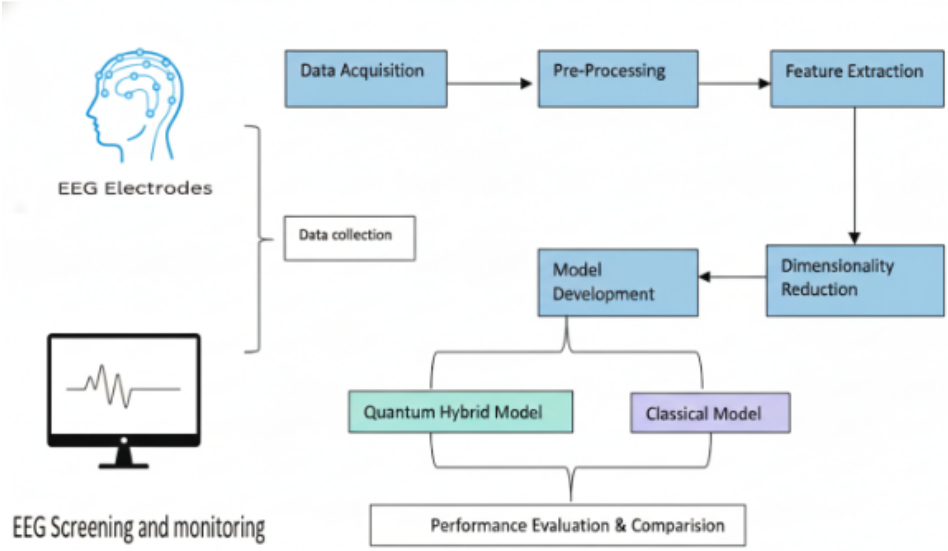


Figure 4.1: Processing pipeline for feature-based seizure state classification.

In this pipeline, EEG recordings obtained from the dataset are first processed through a preprocessing stage to improve signal quality. The signals are then segmented into short analysis windows, from which statistical, spectral, and quantum-inspired features are extracted. These features form the input to machine learning models including classical Support Vector Machines and hybrid quantum–classical classifiers such as Quantum Support Vector Machines [12].

4.2.2 Temporal EEG Modelling Pipeline

While the previous pipeline classifies individual EEG windows, practical seizure monitoring systems must analyse continuous EEG signals and detect seizure progression over time. The temporal modelling pipeline used in this work is shown in Figure 4.2.

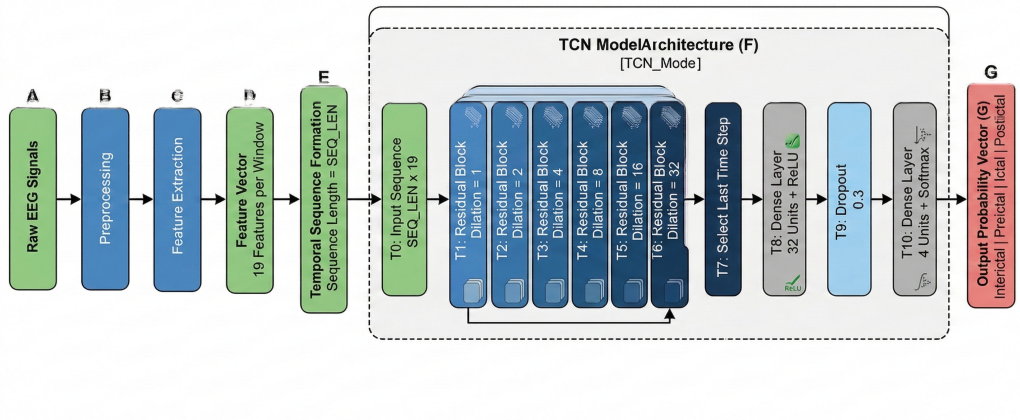


Figure 4.2: Temporal EEG modelling pipeline for seizure stage detection.

In this approach, EEG signals are first preprocessed and segmented into analysis windows. Feature vectors extracted from these windows are then organized into sequences representing continuous neural activity. These sequences are processed using a Temporal Convolutional Network (TCN), which learns temporal relationships between EEG segments and predicts seizure stages over time [21].

4.3 Core System Modules

The proposed EEG analysis system is organized around five core modules that define the major processing stages of the pipeline.

4.3.1 Module 1 – Data Preprocessing

The preprocessing module performs signal conditioning and segmentation operations required for EEG analysis. EEG recordings often contain noise and artifacts caused by physiological activity, measurement hardware, and environmental electrical interference. Filtering techniques are therefore applied to remove low-frequency drift, high-frequency noise, and power-line interference.

Following signal conditioning, the continuous EEG recordings are segmented into fixed-duration analysis windows. These windows represent short segments of neural activity and serve as the basic units for feature extraction.

4.3.2 Module 2 – Feature Extraction and Analysis

The feature extraction module converts preprocessed EEG windows into structured numerical representations that can be used by machine learning models. For each EEG segment, a set of descriptive features is computed to characterize statistical properties, spectral structure, and structural relationships within the signals.

The features used in this work are derived primarily from three domains: time-domain descriptors, frequency-domain descriptors, and quantum-inspired descriptors. These features form the input representation for the machine learning models used in subsequent stages of the system. A detailed description of the feature definitions and mathematical formulations is provided in Chapter 3.

4.3.3 Module 3 – Classical Model Development

The classical model development module constructs machine learning models using the extracted EEG feature representations. These models learn relationships between feature patterns and seizure-related EEG states, enabling classification or prediction tasks to be performed.

4.3.4 Module 4 – Hybrid Quantum-Classical Model Development

The hybrid modelling module incorporates quantum-inspired learning techniques into the analysis pipeline. In this module, classical feature vectors are mapped into a quantum feature space using parameterized quantum circuits. Similarity between samples is then evaluated through quantum kernel methods, enabling nonlinear relationships between features to be captured more effectively [12].

4.3.5 Module 5 – Model Evaluation and Comparison

The model evaluation module measures the performance of the developed models and enables comparison between different modelling strategies. Standard classification metrics

including accuracy, precision, recall, confusion matrices, and receiver operating characteristic curves are used to evaluate window-level classification performance. Event-level evaluation metrics such as false positive rate and preictal detection performance are also considered.

4.4 Quantum-Enhanced Seizure State Classification

The first modelling stage focuses on distinguishing abnormal EEG states associated with seizure activity. Specifically, the classification task involves differentiating between preictal EEG activity and ictal seizure activity. Studying this controlled classification problem allows the discriminative capability of different feature domains to be evaluated, while also providing a controlled setting for examining the effectiveness of quantum-inspired representations and hybrid quantum–classical models before extending the system toward temporal seizure detection.

4.4.1 Data Preprocessing

EEG recordings are first processed using signal conditioning steps to improve signal quality. The raw signals are filtered using a bandpass filter between 0.5 Hz and 40 Hz to remove baseline drift and high-frequency noise. A notch filter is applied to suppress power-line interference, and Independent Component Analysis (ICA) is used to reduce common artifacts such as eye movements and muscle activity.

Following signal conditioning, the EEG recordings are segmented into windows of one second, each representing a short interval of neural activity used for feature extraction and classification. Based on the seizure annotations available in the dataset, windows corresponding to seizure intervals are labeled as ictal. In this framework, preictal activity is defined as EEG segments occurring within the time interval from one to five minutes prior to seizure onset.

4.4.2 Feature Extraction and Representation

For each EEG window, descriptive features are computed to represent the underlying characteristics of the signal. The base feature descriptors used in this work are defined and discussed in detail in Chapter 3. These descriptors are derived from three complemen-

tary domains: time-domain features, frequency-domain features, and quantum-inspired features.

To obtain channel-independent representations suitable for machine learning models, statistical aggregation operations are applied across EEG channels. These aggregations capture global properties of the multichannel signals while reducing dimensionality. As a result, the number of derived features increases compared to the base descriptors defined in Chapter 3, producing a final feature representation consisting of fifty-six unique descriptors. The distribution of features used in this work is summarized in Table 4.1.

Table 4.1: Distribution of feature domains used for seizure state classification.

Feature Domain	Number of Features
Time-domain features	28
Frequency-domain features	17
Quantum-inspired features	11
Total unique features	56

4.4.3 Classical Model Development

Support Vector Machines (SVM) are used as the primary classical classifier due to their strong generalization capability and effectiveness in high-dimensional feature spaces [22]. To evaluate the discriminative power of different feature domains, several feature configurations are examined.

The evaluated configurations include time-domain features only, frequency-domain features only, quantum-inspired features only, a combination of time and frequency features, and the complete feature set containing all domains. Comparing these configurations allows the contribution of each feature domain to seizure state classification to be assessed.

4.4.4 Hybrid Quantum-Classical Model Development

A hybrid quantum–classical classification model is implemented using a Quantum Support Vector Machine (QSVM). In this approach, classical feature vectors are embedded into a

quantum feature space using parameterized quantum circuits. Similarity between samples is computed using a quantum kernel derived from the resulting quantum states [12].

Since quantum circuits scale with the number of input variables, dimensionality reduction is performed prior to quantum encoding. This reduction is achieved through feature selection using Analysis of Variance (ANOVA), which ranks the statistical importance of descriptors. The eight highest ranked features are then selected to form the reduced feature set used for quantum encoding.

In addition, four quantum-inspired descriptors are explicitly included to preserve structural information captured by quantum-based representations. The resulting feature subset therefore contains twelve features in total, corresponding to twelve qubits in the quantum circuit used for kernel computation.

4.4.5 Model Evaluation

The performance of the classical SVM and QSVM models is evaluated using standard classification metrics including accuracy, precision, recall, confusion matrices, and receiver operating characteristic curves.

In addition to these metrics, the improvement obtained using the quantum model is quantified using a quantum gain measure. This metric compares the performance of the QSVM model with the corresponding classical SVM baseline and provides an indication of the benefit obtained from quantum feature mapping.

4.5 Preictal Detection using Temporal Convolutional Networks

While the previous modelling stage focuses on classifying abnormal EEG states, practical seizure monitoring systems must analyse continuous EEG recordings and detect seizure progression over time. In real-world scenarios, seizure stages are not known in advance, and the system must identify gradual changes in neural activity that precede seizure onset.

To address this problem, a temporal EEG modelling approach is developed in which sequences of EEG feature vectors are analysed to learn patterns associated with seizure evolution. The temporal models are trained in a patient-specific manner using EEG recordings from individual patients, allowing the model to learn patient-dependent seizure dynamics.

4.5.1 Data Preprocessing

EEG recordings are first subjected to signal conditioning in order to improve signal quality. The raw signals are filtered using bandpass and notch filters to remove noise and electrical interference while preserving the chronological structure of neural activity. The filtered signals are then segmented into overlapping analysis windows of four seconds with a stride of two seconds, where each window represents a short interval of neural activity and forms the basic unit for feature extraction.

Using seizure annotations provided with the dataset, each EEG segment is assigned to one of four seizure stages: interictal, preictal, ictal, or postictal. The ictal stage corresponds to seizure intervals defined in the dataset annotations. The preictal stage represents the twenty-minute interval preceding seizure onset, while the postictal stage corresponds to the ten-minute recovery period following seizure termination. Since transitions between these stages occur gradually, a fuzzy labeling strategy is applied near stage boundaries by assigning soft labels that distribute probability between adjacent seizure states. To enable temporal modelling, consecutive feature vectors extracted from EEG windows are organized into sequences of length 120, representing several minutes of continuous neural activity and forming the input to the temporal prediction model.

During inference, the interictal and postictal states are treated collectively as a normal background state, since the primary objective of the system is early seizure warning rather than detailed stage discrimination.

4.5.2 Feature Extraction

Each EEG window is represented using a compact feature vector consisting of nineteen descriptors capturing statistical, spectral, and structural properties of the signal. These features summarize relevant characteristics of EEG activity while maintaining a computationally efficient representation suitable for sequential modelling.

The features are derived from three domains: time-domain descriptors that capture waveform dynamics and complexity, frequency-domain descriptors obtained from spectral power estimates, and quantum-inspired descriptors that characterize structural relationships in multichannel EEG signals. The composition of the feature vector used for temporal modelling is summarized in Table 4.2.

4.5.3 Temporal Convolutional Network Model

Temporal modelling of EEG sequences is performed using a Temporal Convolutional Network (TCN). TCN architectures are well suited for sequential biomedical signals because they employ causal convolution layers that ensure predictions depend only on past observations while preserving the chronological order of the data. In addition, dilated convolutions allow the network to capture long-range temporal dependencies in time-series signals without significantly increasing computational complexity [21].

In this work, the network processes sequences of EEG feature vectors of length 120, representing several minutes of continuous neural activity. The TCN architecture consists of stacked convolutional blocks with increasing dilation factors of 1, 2, 4, 8, 16, and 32, enabling the model to learn temporal patterns across multiple time scales. Each block contains causal convolution layers followed by normalization and dropout operations to improve training stability and reduce overfitting. Residual connections are incorporated between convolutional blocks to facilitate gradient flow and allow deeper network architectures.

The final layers of the network map the learned temporal representations to probability estimates for four seizure stages: interictal, preictal, ictal, and postictal. During inference, the interictal and postictal outputs are combined to represent a single normal background state, allowing the system to focus on identifying preictal activity and seizure onset.

4.5.4 Evaluation

Evaluation focuses on event-level performance measures relevant to seizure prediction. These include preictal lead time, which measures how early a seizure warning is generated, and ictal detection latency, which quantifies the delay between seizure onset and detection. Additional metrics include missed preictal warnings, false ictal detections, and the false positive rate (FPR), which evaluates the frequency of incorrect preictal alarms during non-seizure periods.

4.6 Software and Hardware Requirements

The proposed system is implemented in Python using several scientific computing and machine learning libraries, including NumPy, SciPy, Pandas, Scikit-learn, TensorFlow,

and PennyLane. Signal processing and feature extraction procedures are implemented using numerical computing libraries, while classical machine learning models are developed using Scikit-learn. The temporal deep learning model is implemented using TensorFlow.

Quantum kernel simulations used for the QSVM model are implemented using the PennyLane quantum computing framework. The experiments are conducted on standard computing hardware, with GPU acceleration utilized for training the deep learning models.

A web-based interface developed using React is used to visualize EEG signals and model predictions generated by the seizure monitoring system.

4.7 User Interface

A web-based interface was developed to demonstrate the proposed seizure monitoring system. The interface provides a visual environment for observing EEG signals and model predictions in a simulated real-time setting. Predictions generated by the Temporal Convolutional Network (TCN) model are streamed sequentially, enabling visualization of temporal neural activity and predicted seizure stages.

The interface consists of a landing page that introduces the system and provides access to the monitoring dashboard. The dashboard displays EEG signal plots along with extracted feature statistics and predicted neural states generated by the TCN model. These visual components allow users to observe the temporal evolution of EEG activity and corresponding seizure stage predictions.

4.8 Summary

This chapter presented the design of the proposed EEG-based seizure state classification and prediction system. The system is organized around a modular architecture consisting of preprocessing, feature extraction, model development, and evaluation components.

Two complementary modelling approaches are implemented within this system. The first approach investigates feature-based seizure state classification using classical and hybrid quantum–classical models. The second approach extends the system toward temporal EEG modelling, enabling prediction of seizure stages and identification of preictal activity from continuous EEG recordings.

Table 4.2: Feature set used for temporal EEG modelling (19 features).

Feature Domain	Feature	Description
Frequency-domain	θ/δ	Ratio of theta to delta band power
Frequency-domain	α/θ	Ratio of alpha to theta band power
Frequency-domain	β/α	Ratio of beta to alpha band power
Frequency-domain	γ/β	Ratio of gamma to beta band power
Frequency-domain	δ/α	Ratio of delta to alpha band power
Frequency-domain	δ/β	Ratio of delta to beta band power
Frequency-domain	θ/β	Ratio of theta to beta band power
Frequency-domain	Relative delta power	Fraction of total spectral power in delta band
Frequency-domain	Relative theta power	Fraction of total spectral power in theta band
Time-domain	Hjorth activity	Signal variance
Time-domain	Hjorth mobility	Estimate of mean signal frequency
Time-domain	Hjorth complexity	Measure of waveform complexity
Time-domain	Higuchi fractal dimension ($k = 6$)	Short-scale fractal dimension
Time-domain	Higuchi fractal dimension ($k = 50$)	Long-scale fractal dimension
Time-domain	Wigner skewness	Time–frequency asymmetry measure
Quantum-inspired	Von Neumann entropy	Entropy of density matrix representation
Quantum-inspired	Entanglement entropy	Partial-system entropy
Quantum-inspired	Quantum Fisher information	Temporal change in system state
Quantum-inspired	Energy change	Temporal variation in signal energy

Chapter 5

Results and Discussion

This chapter presents the experimental results obtained from the proposed EEG seizure analysis system. Prior to the development of the final framework, extensive experimentation was conducted with various feature representations and machine learning models to analyze EEG signals and identify effective approaches for seizure detection. Based on the insights gained from this exploration, the final system developed in this work consists of two main components. The first component focuses on feature-based seizure state classification using classical and hybrid quantum–classical models. The second component extends the system toward temporal EEG modelling using a Temporal Convolutional Network (TCN) to detect seizure progression and provide early warning of seizure onset.

5.1 Feature-Based Seizure State Classification

The first stage evaluates the ability of engineered EEG features to distinguish between preictal and ictal brain states. Multiple feature domains were investigated including time-domain, frequency-domain, and quantum-inspired descriptors.

Figures 5.1 and 5.2 present the confusion matrices and ROC curves obtained for the different feature configurations. Figure 5.1 illustrates the performance of the classical SVM classifier when trained using time-domain features, frequency-domain features, and the combined time–frequency feature set. These feature representations capture complementary characteristics of EEG signals including waveform dynamics and spectral power distribution.

Figure 5.2 shows the results obtained using quantum-inspired descriptors and the complete combined feature set. Quantum-inspired features capture structural relationships between EEG channels and provide additional information about underlying neural activity patterns.

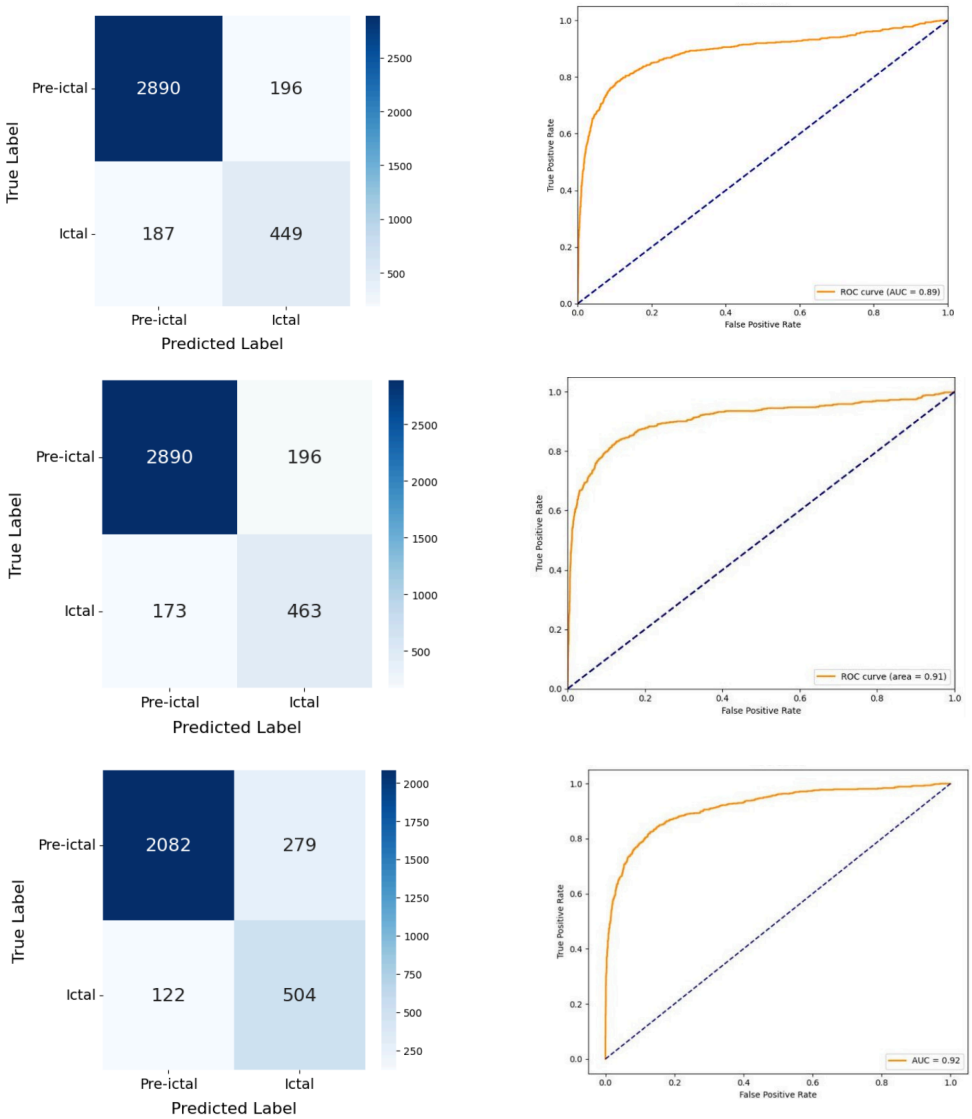


Figure 5.1: Confusion matrices and ROC curves for classical SVM models using (top) time-domain features, (middle) frequency-domain features, and (bottom) combined time–frequency features.

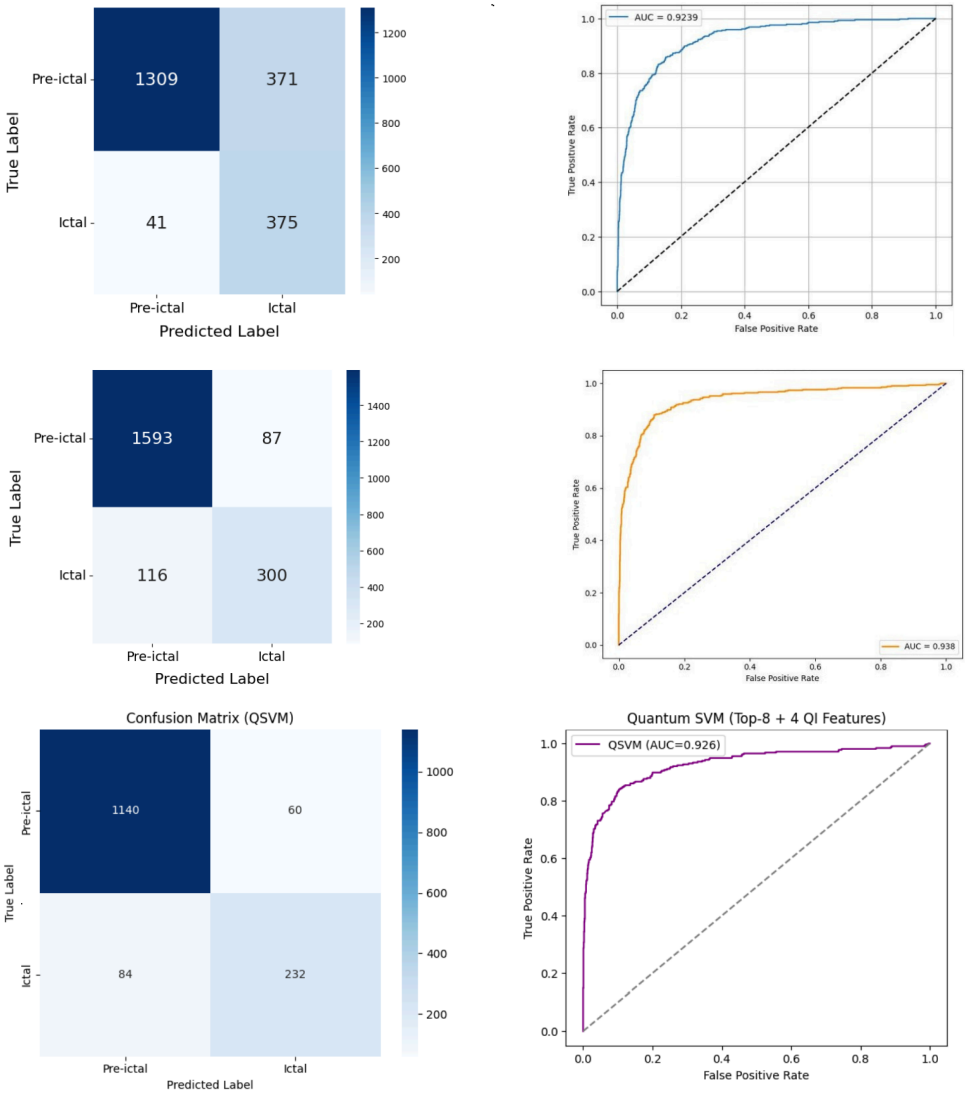


Figure 5.2: Confusion matrices and ROC curves for SVM models using (top) quantum-inspired features and (middle) the complete combined feature set and (bottom) hybrid quantum–classical QSVM model.

Table 5.1: Performance comparison of different EEG feature representations.

Feature Set	Preictal		Ictal		ROC-AUC
	Precision	Recall	Precision	Recall	
Time-domain only	0.94	0.94	0.70	0.71	0.889
Frequency-domain only	0.94	0.94	0.70	0.73	0.9078
Time + Frequency	0.94	0.88	0.64	0.81	0.9151
Quantum-inspired only	0.97	0.78	0.50	0.90	0.9239
All features combined	**0.93**	**0.95**	**0.78**	**0.72**	**0.938**

Table 5.1 summarizes the classification performance obtained using different feature configurations. The results show that combining multiple feature domains improves classification performance compared to using individual feature groups alone. Time-domain descriptors capture waveform characteristics, while frequency-domain features describe spectral power distribution in EEG signals. When these feature domains are combined, the classifier is able to utilize complementary information from multiple signal representations.

The best classical performance was achieved using the complete feature set, which obtained strong precision and recall values for both seizure states. The overall classification accuracy of the classical SVM model was approximately **90%**.

5.2 Hybrid Quantum–Classical Model

To investigate the potential benefits of quantum-enhanced learning, a hybrid quantum–classical model was implemented using a Quantum Support Vector Machine (QSVM). In this approach, classical feature vectors are mapped into a quantum feature space using parameterized quantum circuits, allowing nonlinear relationships between EEG features to be captured through quantum kernel computation.

Feature selection using ANOVA was applied to identify the most informative descriptors. The final quantum model used eight classical features together with four quantum-inspired descriptors. The classification performance of the QSVM model is summarized in Table 5.2.

Table 5.2: Performance of the hybrid quantum–classical model (QSVM).

Class	Precision	Recall	F1-score
Preictal	0.93	0.95	0.94
Ictal	0.79	0.73	0.76

Table 5.3: Comparison between classical SVM and hybrid quantum–classical model.

Model	Features Used	Accuracy	Ictal F1-score
Classical SVM (combined features)	56	0.90	0.75
Hybrid Quantum SVM	12	0.91	0.76

The confusion matrix and ROC curve corresponding to the hybrid quantum model are illustrated in Figure 5.2.

5.3 Classical vs Hybrid Quantum Model Comparison

Table 5.3 compares the best classical model with the hybrid quantum–classical model. The classical SVM model uses the complete set of 56 engineered features, while the QSVM model operates on a reduced subset of 12 features selected through ANOVA and quantum-inspired feature inclusion.

Although the hybrid model achieved similar classification performance, the quantum kernel computation introduces additional computational overhead compared to the classical SVM approach. As a result, while quantum-inspired approaches demonstrate promising representational capabilities, classical models remain more computationally efficient for practical deployment.

5.4 Temporal Seizure Prediction using TCN

While the previous experiments focused on classification of individual EEG windows, real-world seizure monitoring requires analysis of continuous EEG recordings in order to detect seizure progression over time. To address this problem, a Temporal Convolutional Network (TCN) was implemented to analyse sequences of EEG feature vectors extracted

Table 5.4: Event-level seizure detection results for patient CHB08 using the TCN model

Seizure Event	True Time (s)	Ictal Detected (s)	Preictal Warning (s)	**Lead Time (s)**
Seizure 1 (Training)	2670	2680	278	**2392**
Seizure 2 (Training)	2856	2868	2040	**816**
Seizure 3 (Validation)	2988	3002	2724	**264**
Seizure 4 (Unseen)	2417	2430	898	**1519**
Seizure 5 (Unseen)	2083	2086	1646	**437**

Table 5.5: Event-level seizure detection results for patient CHB10 using the TCN model

Seizure Event	True Time (s)	Ictal Detected (s)	Preictal Warning (s)	**Lead Time (s)**
Seizure 1 (Training)	6313	6314	5212	**1101**
Seizure 2 (Training)	6888	6860	5942	**946**
Seizure 3 (Validation)	2382	2384	2112	**270**
Seizure 4 (Unseen)	3021	3040	2146	**875**
Seizure 5 (Unseen)	3801	3818	3008	**793**
Seizure 6 (Unseen)	4618	4634	238	**4380**
Seizure 7 (Unseen)	1383	1388	238	**1145**

from overlapping EEG windows.

The model processes windows of four seconds with a stride of two seconds, enabling the network to capture temporal patterns that emerge prior to seizure onset. The network predicts probabilities corresponding to seizure stages including interictal, preictal, ictal, and postictal states. For the purpose of early warning analysis, interictal and postictal predictions are grouped together as normal activity.

5.5 Event-Level Seizure Detection Results

The temporal prediction system was evaluated at the seizure-event level. Tables 5.4 and 5.5 summarize the seizure detection results obtained for patients CHB08 and CHB10 respectively. These tables report the true seizure onset time, the time at which the ictal state was detected by the model, the time at which a preictal warning was generated, and the resulting preictal lead time.

Table 5.6 summarizes the overall event-level performance of the proposed TCN-based

Table 5.6: Summary of event-level seizure prediction performance of the TCN model

Patient	Avg Preictal Lead Time (s)	Avg Ictal Detection Latency (s)	False Ictal Alarms	False Positive Rate (per hour)
CHB08	1086	10.4	4	0.05
CHB10	1359	4.6	0	0.43

seizure prediction system. The model successfully generated preictal warnings prior to seizure onset for all evaluated seizure events across both patients.

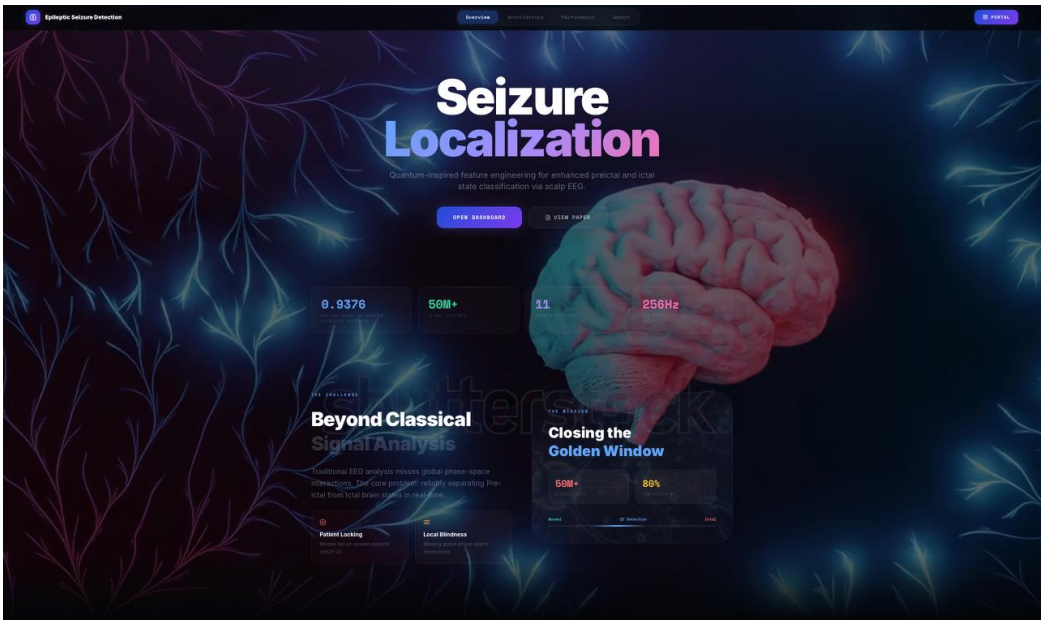
5.6 User Interface Demonstration

Figures 5.3a and 5.3b illustrate the web-based interface developed to demonstrate the seizure monitoring system. The interface presents EEG signals and predicted seizure stages generated by the Temporal Convolutional Network model in a simulated real-time monitoring environment.

5.7 Discussion

The experimental results demonstrate that engineered EEG features combined with machine learning models are capable of capturing discriminative patterns associated with seizure activity. The results presented in Table 5.1 show that the performance of the classical SVM classifier improves when multiple feature domains are combined. Time-domain descriptors capture signal amplitude variations and waveform dynamics, while frequency-domain features characterize the spectral power distribution of EEG signals. When these representations are used together, the classifier is able to utilize complementary information from both domains, resulting in improved classification performance.

Quantum-inspired features were also evaluated as an alternative feature representation. As shown in Figures 5.2 and Table 5.1, these features achieved competitive performance even when used independently. This suggests that structural information derived from multichannel EEG relationships can provide useful discriminative information for seizure state classification. However, although the hybrid quantum–classical QSVM model achieved performance comparable to the best classical SVM model (Table 5.3), the experiments indicate that the quantum kernel computation introduces significantly higher computational cost. As a result, the practical advantage of the hybrid model in



(a) Landing page of the seizure monitoring interface.



(b) Monitoring dashboard displaying EEG signals and predicted neural states.

Figure 5.3: Web-based interface developed to demonstrate the seizure monitoring system.

this study lies primarily in the ability to achieve similar performance using a reduced feature set rather than in improved classification accuracy.

While the feature-based classification models achieved strong performance for distinguishing preictal and ictal EEG segments, the results also highlight an important limitation of window-level classification approaches. These models operate on isolated EEG segments and therefore do not directly capture temporal relationships between consecutive EEG windows. In real-world seizure monitoring scenarios, seizure onset typically evolves gradually over time, and early prediction requires the ability to analyse temporal patterns in continuous EEG signals.

To address this limitation, the Temporal Convolutional Network (TCN) model was evaluated using sequential EEG feature inputs. The event-level results shown in Tables 5.4 and 5.5, together with the summary metrics in Table 5.6, indicate that the model was able to generate preictal warnings prior to seizure onset for all evaluated seizure events. The average preictal lead time exceeded 16 minutes for both patients, demonstrating the ability of the temporal model to detect early changes in neural activity before seizure onset.

The event-level analysis also shows that ictal detection latency remained relatively small across the evaluated seizures, indicating that the system was able to rapidly identify seizure onset once ictal activity emerged. In addition, a low number of false ictal alarms was observed in the evaluated recordings, and a low false positive rate during the test sequences.

Chapter 6

Conclusion and Future Scope

This project investigated the development of an EEG-based seizure analysis system capable of identifying seizure-related brain activity and providing early warning of seizure onset. Epileptic seizures often occur unpredictably and can significantly affect patient safety and quality of life. Reliable seizure monitoring and prediction systems therefore have the potential to assist clinicians and patients by providing timely alerts and enabling preventive actions. Motivated by this challenge, this work explored machine learning and quantum–classical approaches for analysing EEG signals and identifying patterns associated with epileptic seizures.

The proposed system was implemented as a structured EEG analysis pipeline consisting of preprocessing, feature extraction, model development, and evaluation stages. Raw EEG recordings from the CHB-MIT scalp EEG dataset were first processed using signal conditioning techniques to reduce noise and electrical interference. The recordings were then segmented into analysis windows, from which descriptive features capturing both temporal and spectral characteristics of neural activity were extracted.

The first stage of the study focused on feature-based seizure state classification. Classical Support Vector Machine (SVM) models were trained using different feature combinations in order to evaluate their ability to distinguish between preictal and ictal EEG activity. The results showed that combining multiple feature domains improved classification performance compared to using individual feature groups alone. The best classical model achieved an overall classification accuracy of approximately 90%, demonstrating that engineered EEG features can effectively capture seizure-related patterns in neural activity.

In addition to classical machine learning models, a hybrid quantum–classical approach was also explored using a Quantum Support Vector Machine (QSVM). In this approach, classical features were embedded into a quantum feature space using parameterized quan-

tum circuits. The hybrid model achieved performance comparable to the best classical model while using a reduced feature set. However, the experiments also indicated that quantum kernel computation introduces considerable computational overhead when implemented on classical simulators. As a result, while hybrid quantum–classical models demonstrate interesting representational capabilities, classical models remain more practical under current computational constraints.

Although window-level classification provides useful insight into EEG feature representations, practical seizure monitoring requires the ability to analyse continuous EEG recordings. To address this requirement, the second stage of the study focused on temporal EEG modelling. A Temporal Convolutional Network (TCN) was implemented to analyse sequences of EEG feature vectors extracted from continuous recordings. By modelling temporal dependencies between consecutive EEG segments, the TCN was able to capture gradual changes in neural activity that occur prior to seizure onset.

Event-level evaluation demonstrated that the temporal model was capable of generating early warnings before seizure onset for the evaluated patients. The average seizure prediction horizon exceeded 16 minutes, indicating that the system was able to detect preictal patterns well before ictal activity began. In addition, the model showed low ictal detection latency once seizures started. These results demonstrate the feasibility of developing an EEG-based seizure early warning system using temporal machine learning models. The current implementation, however, is patient-specific, meaning that the model is trained and evaluated individually for each patient.

Despite these encouraging results, several limitations remain. EEG signals are inherently noisy and complex, and residual artifacts may still remain even after preprocessing. Such noise can influence feature extraction and model predictions, highlighting the importance of more robust preprocessing and artifact removal techniques. Furthermore, the evaluation in this work was conducted on a limited number of patients and seizure events, and broader validation across larger datasets would be necessary to fully assess the reliability of the proposed approach.

Future work can therefore focus on improving signal preprocessing methods to further reduce the impact of noise in EEG recordings and conducting more extensive evaluations across larger patient cohorts. Another promising direction involves the continued exploration of hybrid quantum–classical approaches. While the current implementation

relies on quantum circuit simulations, future advances in practical quantum hardware may enable more efficient implementation of quantum kernel methods and provide new opportunities for analysing high-dimensional biomedical signals such as EEG.

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List of Publications

1. S. Izudheen, F. D. Achandy, J. G. Joseph, C. Pradeep, and J. Chemmanoor, "Epileptic Seizure State Classification Using Quantum Inspired Techniques", presented at the *International Conference on Methods in Data Science*, Mar Athanasius College (Autonomous), Kerala, India, 2026. Under review for publication in the *Springer Proceedings in Mathematics & Statistics*.

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions for complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design or develop systems, components or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and the World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion, and adhere to national and international laws.

PO8: Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse or multidisciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, including the ability to comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for (i) independent and life-long learning, (ii) adaptability to new and emerging technologies, and (iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

COURSE OUTCOMES (CO):

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology or research gaps and propose innovative or creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

Appendix C: CO-PO-PSO Mapping

CO - PO - PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

Table 6.1: Justifications for CO-PO Mapping

Mapping	**Level**	**Justification**
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.

Mapping	Level	Justification
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.

Mapping	Level	Justification
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.

Mapping	Level	Justification
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.